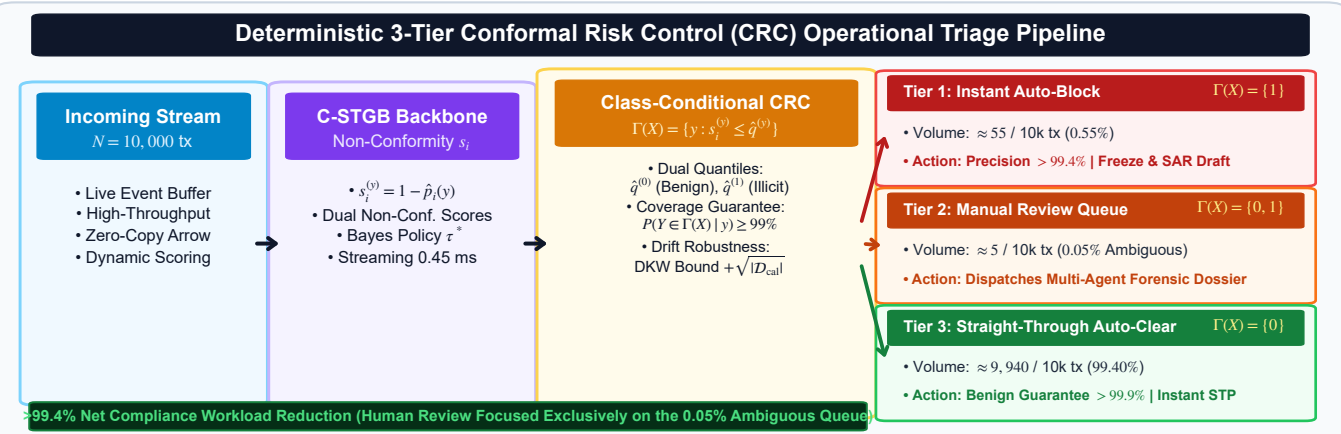

Deterministic 3-Tier Conformal Risk Control (CRC) Operational Triage Pipeline
Incoming Stream
$N = 10,000$ tx
• Live Event Buffer
• High-Throughput
• Zero-Copy Arrow
• Dynamic Scoring
C-STGB Backbone
Non-Conformity s_i
• $s_i^{(y)} = 1 - \hat{p}_i(y)$
• Dual Non-Conf. Scores
• Bayes Policy τ^*
• Streaming 0.45 ms
Class-Conditional CRC
$\Gamma(X) = \{y : s_i^{(y)} \le \hat{q}^{(y)}\}$
• Dual Quantiles:
$\hat{q}^{(0)}$ (Benign), $\hat{q}^{(1)}$ (Illicit)
• Coverage Guarantee:
$P(Y \in \Gamma(X) \mid y) \ge 99\%$
• Drift Robustness:
DKW Bound $+\sqrt{|D_{\text{cal}}|}$
Tier 1: Instant Auto-Block
$\Gamma(X) = \{1\}$
• Volume: ≈ 55 / 10k tx (0.55%)
• Action: Precision $> 99.4\%$ | Freeze & SAR Draft
Tier 2: Manual Review Queue
$\Gamma(X) = \{0, 1\}$
• Volume: ≈ 5 / 10k tx (0.05% Ambiguous)
• Action: Dispatches Multi-Agent Forensic Dossier
Tier 3: Straight-Through Auto-Clear
$\Gamma(X) = \{0\}$
• Volume: $\approx 9,940$ / 10k tx (99.40%)
• Action: Benign Guarantee $> 99.9\%$ | Instant STP
>99.4% Net Compliance Workload Reduction (Human Review Focused Exclusively on the 0.05% Ambiguous Queue)